National University of Singapore
School of Computing

BT2102 Data Management and Visualisation

ASSIGNMENT 3 (GROUP PROJECT): DATA VISUALISATION AND STORYTELLING

Total marks: 35

DATASET PROVIDED

You are provided with the dataset "**BeijingHousePrices.xlsx**" on the housing prices in Beijing. The data is taken from Lianjia.com and it spans the period from 2002 to 2018. A download of the data (available in https://www.kaggle.com/ruiqurm/lianjia) is provided in the Assignment 3 folder. All the data can be fetched from https://bj.lianjia.com/chengjiao and you may use them where appropriate.

Dataset Details

The dataset and its description are available in Appendix A.

WHAT ARE YOU REQUIRED TO DO?

Your team is the data analytics team of a large organization with capabilities in real estate investment, development, leases, and mall management. Your management has interest in venturing into Beijing properties. They provided your team with the dataset as given here and would like your team to answer this level-1 question **"What is the best investment strategy to maximize returns in Beijing properties?"**

You may approach this presentation in this manner:

1. Consider the above **level-1 business question**.
2. Add in any further lower-level questions for producing the appropriate insights for your story.
3. Conduct appropriate data cleaning.
4. Conduct exploratory and explanatory analysis featuring appropriate *representation* and *presentation* of data**.**
5. Use Tableau, MS Excel, Python packages or any tool of your choice to produce your data visualisations.
6. Prepare your presentation with the data visualisations into a data story and produce a **coherent presentation** with recommendations for your story.

You may assume that the data is recent enough for your analysis and recommendations.

Note that data insights lead to the production of story, but they are not stories themselves. Your team may seek out other ***relevant*** data to supplement the currently available dataset in your presentation.

Copyright © Danny Poo, 2026

DELIVERABLES

1. A PowerPoint (or similar) storytelling presentation.
    a. There is no restriction on the number of slides but note that your group has only 10 minutes for the presentation.

SUBMISSION

1. Zip the PowerPoint file and name it as "**Team_XX_A3.zip**" (for example, file name for group 9 is "Team_09_A3.zip" and for group 18 it is "Team_18_A3.zip"). **PLEASE FOLLOW THIS NAMING CONVENTION ACCORDINGLY.**
2. Upload the zip file to "Assignment 3" folder in Canvas->Assignments by the assignment due date.
3. **EACH MEMBER OF THE TEAM MUST SUBMIT A COPY OF THE SAME TEAM ZIP FILE TO THE CANVAS ASSIGNMENT FOLDER.**
4. Please be reminded that plagiarism is a serious offence. Disciplinary actions will be taken against those caught cheating.

PROJECT CONSULTATION

1. Due to the large class size, Mr Huang Junjing (my graduate tutor) and I will answer your questions in project consultation. Contact/email Mr Huang Junjing (junjing@nus.edu.sg), or me (dannypoo@nus.edu.sg) should you require consultation.

EVALUATION

1. Each group is given 10 minutes to present and 5 minutes Questions and Answers during the allocated 15 minutes. All presentations will be conducted online via Zoom Meetings.
2. Evaluation timeslot will be scheduled nearer week 12.
3. Your group will be evaluated on the data visualisations and storytelling presentation.
4. Each member in the group is required to present, even for a minute.
5. Evaluation criteria:
    a. **Business Questions to Answer (5 marks):**
        i. Define very clear, important, and interesting business questions that can be answered by visualizations.
    b. **Knowledge/Understanding (4 marks):**
        i. Demonstrate full knowledge of the subject of data visualization and dataset.
        ii. Understanding of the dataset.
        iii. Accurate use and perspective of the dataset.
    c. **Story (6 marks):**
        i. A convincing, coherent story crafted from data insights.
    d. **Visualization (5 marks):**
        i. Visualization diagrams are used professionally to give informative presentations.
    e. **Layout (4 marks):**
        i. **Colors**: Use of different and appropriate color of each diagram.
        ii. **Neatness**: Very neat and clearly readable diagrams.
        iii. **Title and accurate labels**: Clear title for the charts. Labels located correctly. No spelling errors.
    f. **Creativity (6 marks):**
        i. Synthesizing data analysis ideas to reach creative insights such as the use of predictive analytics, clustering, and other data analytics techniques to derive insights for the presentation.
    g. **Presentation style (2 marks):**
        i. Professional presentation by presenters.
        ii. Appropriate dressing
    h. **Questions and Answers (3 marks):**
        i. Ability to respond to questions posed.

Copyright © Danny Poo, 2026

Appendix A

FIELD	DESCRIPTION
url	url of property (house)
id	Id of transaction
Lng	Longitude of property
Lat	Latitude of property
Cid	Community id
tradeTime	Transaction date
DOM	Active days on market. More details in https://en.wikipedia.org/wiki/Days_on_market
followers	The number of people following/interested in the transaction
totalPrice	The total price of property (in Chinese Yuan and in ten thousands)
price	The average price (in Chinese Yuan) per square meters
square	The size of property (in square meters)
livingRoom	The number of bedroom
drawingRoom	The number of living room
kitchen	The number of kitchen
bathroom	The number of bathroom
floor	The number of floors in property. "High", "Middle", "Low" are indication of the level of floor for that house unit. E.g., "High 26" means there are 26 floors in that building and the unit is located on a high floor i.e., within 19th to 26th floor.
buildingType	1:tower, 2:bungalow, 3:combination of plate and tower, 4:plate
constructionTime	When was the property constructed
renovationCondition	1:other, 2:rough, 3:simplicity, 4:refined decoration
buildStructure	1:unknown, 2:mixed, 3:brick and wood, 4: brick and concrete, 5:steel, 6:steel-concrete composite
ladderRatio	the proportion between number of residents on the same floor and number of ladders. It describes how many ladders a resident has on average.
elevator	0:does not have elevator, 1:has elevator
fiveYearsProperty	0:policy does not apply, 1: policy applies. Chinese government housing policy: if a person has owned a property for over five years before selling it AND if the property is the only real estate he owns in Beijing, then the Value Added Tax on this transaction can be waived.
subway	0:No subway nearby, 1:subway nearby
district	1:Dongcheng District, 2:Fengtai District, 3:Tongzhou District, 4:Daxing District, 5:Fangshan District, 6:Changping District, 7:Chaoyang District, 8:Haidian District, 9:Shijingshan District, 10:Xicheng District, 11:Pinggu District, 12:Mentougou District, 13:Shunyi District. 3 districts are not included in dataset: Yanqing district, Huairou district, and Miyun district
communityAverage	The number of people living in the community

Copyright © Danny Poo, 2026